Technische Universität Dortmund
Fakultät Statistik

Visualisierung komplexer Datenstrukturen
Final Report

Visualization of Vancomycin Data

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1. Introduction

Vancomycin is a potent glycopeptide antibiotic commonly used to treat life-threatening infections caused by Gram-positive bacteria, including methicillin-resistant *Staphylococcus aureus* (MRSA). Its therapeutic management requires precise monitoring due to a narrow therapeutic window: overdosing increases the risk of nephrotoxicity, while underdosing may contribute to antimicrobial resistance. Key clinical variables—such as kidney function, illness severity, and bacterial sensitivity—can significantly impact Vancomycin pharmacokinetics and dosing decisions.

This report aims to explore and visually analyze Vancomycin therapy using clinical data collected from Cologne-Merheim Hospital between 2010 and 2022. The focus lies on identifying relevant clinical patterns through the use of descriptive statistics and visualizations, emphasizing the interplay between Vancomycin levels, renal function, illness severity, and bacterial resistance.

The analysis includes time-dependent trends of kidney function (eGFR), serum Vancomycin levels, the effect of illness severity (SAPS and SOFA scores), and the influence of nephrotoxic medications. In doing so, the study could possibly offers insights into safer and more individualized use of Vancomycin in hospital settings.

2. Problem Statement

This project investigates the influence of comorbidities, microbiological findings, kidney function, and illness severity on Vancomycin therapy in critically ill patients. The aim is to identify risk factors for under- or overdosing and evaluate the need for personalized dosing strategies.

2.1. Data Description and Quality

The dataset originates from a retrospective cohort of 922 ICU patients treated with continuous Vancomycin infusion between May 2010 and August 2022 at Krankenhaus Köln-Merheim. It includes 63 variables covering patient demographics, renal function, comorbidities, illness severity, medication use, microbiological findings, and Vancomycin therapy.

Demographics include categorical variables such as gender and dates (birth, death), and continuous variables like height and weight. Renal function is measured by serum creatinine (SCr) and estimated glomerular filtration rate (eGFR) at baseline, 24h, 48h, 72h, and end of therapy. Comorbidities (e.g., cardiovascular disease, chronic kidney disease, diabetes, malignancy) are coded as binary variables. Illness severity is assessed via SAPS and SOFA scores (ordinal), supported by markers such as leukocytes and CRP. Nephrotoxic drugs (e.g., ACE inhibitors, ARBs, aminoglycosides, loop diuretics, NSAIDs, PipTaz, vasopressors) are also recorded as binary indicators.

Infection-related variables include flags for sepsis, shock, endocarditis, catheter infections, pneumonia, and SSTIs. Microbiological results (MRSA, MSSA, Enterococcus, Enterobacterales, fungi) are similarly binary.

Vancomycin therapy is documented via continuous variables for loading dose (LD), maintenance doses (MD24, MD48, MD72), and serum concentrations (C24, C48, C72). Therapy start/end dates are ordinal, and patient age at therapy start was derived.

Missing values appear in key variables: SOFA score (162 cases), Vancomycin C72 (96), MD72 (10), SCr72 and eGFR72 (9 each), and mortality date (697). Relevant analyses excluded cases with missing data to maintain validity.

2.2. Objectives of the Project

The primary objective of this project is to explore the clinical determinants and outcomes of Vancomycin therapy through data visualization, with a focus on renal function, illness severity, and bacterial resistance. Visual analyses were used to identify patterns that support safer and more effective treatment decisions.

Key research questions include:

1. How does illness severity affect serum Vancomycin concentrations and kidney function? This analysis investigates whether higher SAPS or SOFA scores are associated with increased Vancomycin accumulation and declining eGFR, indicating compromised renal clearance in critically ill patients.

2. Do Vancomycin levels differ between infections with resistant versus sensitive bacteria? By comparing serum levels in patients based on microbiological culture results, the study evaluates whether resistance status influences dosing and drug exposure.

3. How does the use of nephrotoxic drugs impact renal function? This explores associations between common nephrotoxic medications (e.g., loop diuretics, ACEI, vasopressors) and elevated serum creatinine, stratified by gender and drug combinations.

These objectives were addressed using a combination of statistical summaries and interpretable graphics, allowing for clinical insights into Vancomycin dosing practices and renal risk management.

3. Analysis and Interpretation

This chapter explores key patterns in Vancomycin use through visual analysis. Each subsection focuses on a specific research question supported by graphical interpretation.

3.1. Influence of Illness Severity on Kidney Function

To investigate the effect of illness severity on renal function, we analyzed the estimated glomerular filtration rate (eGFR) across five timepoints: at the start of Vancomycin therapy (eGFRStart), and after 24, 48, and 72 hours, as well as at the end of treatment (eGFREnd). eGFR serves as a clinical indicator of kidney function, where values below 90 mL/min/1.73 m^2 are generally considered impaired.

Severity of illness was assessed using two clinical scoring systems: the Simplified Acute Physiology Score (SAPS) and the Sepsis-related Organ Failure Assessment (SOFA) Score. To enable meaningful comparison of patients based on illness severity, both SAPS and SOFA scores were divided into three groups using quantile-based thresholds. Specifically, we calculated the 33rd and 66th percentiles for each score using the R `quantile()` function. Values below the 33rd percentile were classified as low severity, those between the 33rd and 66th percentiles as medium severity, and values above the 66th percentile as high severity. This method ensures a balanced distribution of patients across the severity levels, allowing for clearer group comparisons.

Figure 1 displays the development of eGFR over time, stratified by SAPS and SOFA severity groups. The line and ribbon plot was created using `ggplot2` in R and the complete code can be found in Listing 2 in the Appendix.

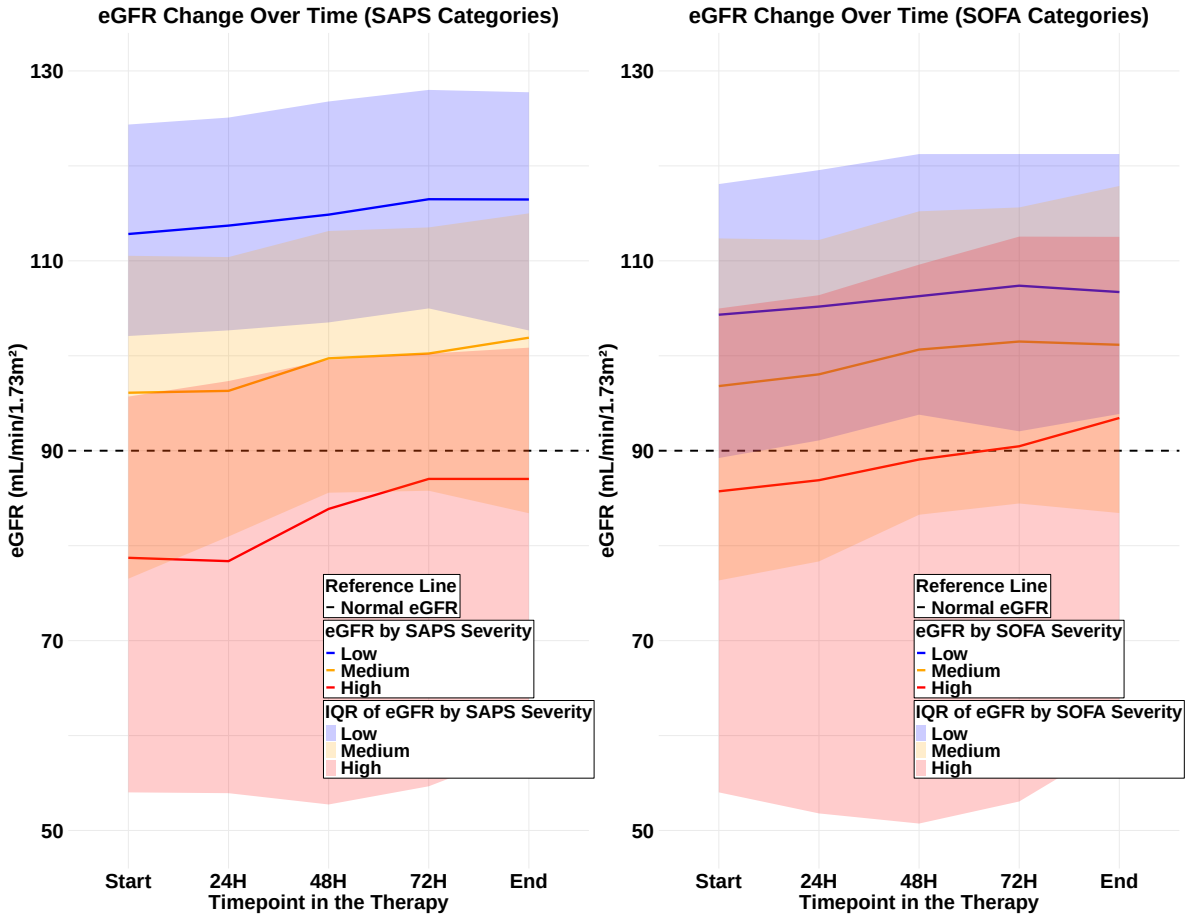

Figure 1: eGFR (mL/min/1.73 m^2) Change Over Time by SAPS and SOFA Categories. The dashed line marks the threshold for normal kidney function (90 mL/min/1.73 m^2). The colored bands represent the interquartile range (IQR).

Patients classified as severely ill (high SAPS or SOFA scores) exhibited substantially lower eGFR values over time, indicating a progressive decline in renal function. In contrast, patients in the low SAPS and SOFA categories generally maintained stable and near-normal kidney performance. Notably, the SAPS-based stratification shows clearer separation between groups than SOFA, suggesting greater discriminatory power in detecting renal risk. Moreover, the interquartile range bands widen over time in patients with higher severity, indicating increased variability in renal outcomes. In both score categories, the **low and medium SAPS and SOFA groups** maintain eGFR near and above the normal levels. In SOFA Category, the patient with **high SOFA** scores tends to recover at the end of the therapy, indicated by the median line crossing over the 90 mL/min/1.73 m^2 threshold.

3.2. Influence of Illness Severity on Vancomycin Concentrations

To examine how illness severity affects Vancomycin pharmacokinetics, we analyzed serum Vancomycin concentrations measured at 24, 48, and 72 hours after treatment initiation (C24, C48, and C72). The dataset included SAPS and SOFA scores for each patient, which were previously categorized into low, medium, and high severity groups based on quantile thresholds. Horizontal lines and the values inside the boxplot represent the median.

Figure 2 illustrates the distribution of Vancomycin concentrations across these severity groups. The boxplots were generated in R using a custom plotting function and `ggplot2`, and the code is available in Listing 3 in the Appendix.

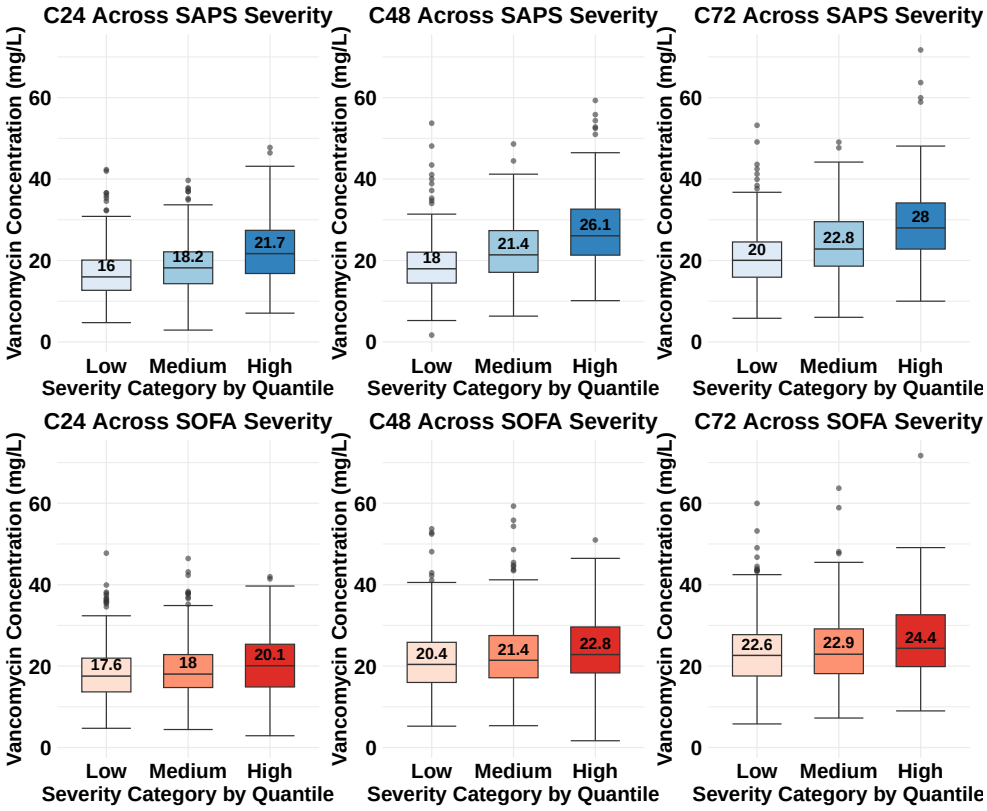

Figure 2: Vancomycin Serum Concentration (mg/L) at 24H, 48H, and 72H stratified by SAPS and SOFA Severity Levels.

Patients in the high SAPS group consistently displayed elevated Vancomycin concentrations at all timepoints. This pattern is consistent with impaired renal clearance in critically ill individuals. Furthermore, the SAPS-based classification provided clearer

group differentiation than SOFA, which exhibited more overlap among categories. In both stratifications, Vancomycin levels remained relatively stable over time, possibly due to the use of continuous infusion protocols. However, the persistently high levels raise concerns about potential drug accumulation and the necessity of individualized dosing to avoid toxicity.

3.3. Vancomycin Serum Levels by Bacterial Sensitivity

Vancomycin dosing often depends on the susceptibility of the infecting organism. Patients were grouped based on bacterial culture results into either "Sensitive," where the pathogen was susceptible to Vancomycin, or "Resistant," where decreased susceptibility was detected. Vancomycin serum levels were measured at 24, 48, and 72 hours, and the data were reshaped into long format to enable visualization via violin plots. These plots illustrate both the distribution and central tendency of concentrations over time.

Figure 3 presents these distributions. The violin and boxplots were generated using `ggplot2` and reshaped data via `reshape2`, with the full code shown in Listing 4.

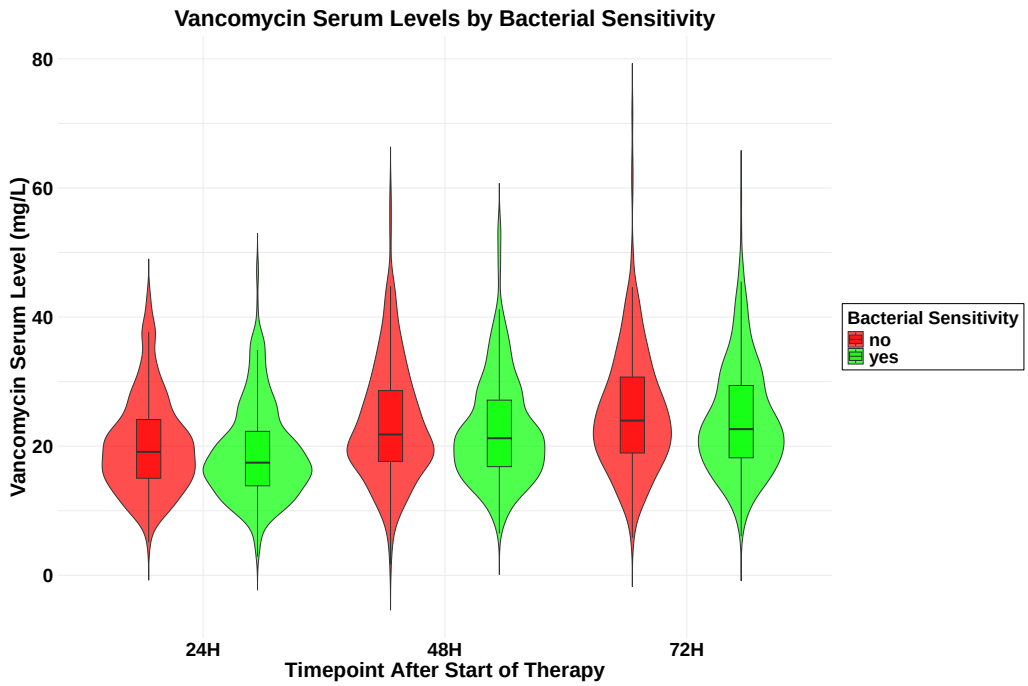

Figure 3: Distribution of Vancomycin Serum Levels by Bacterial Sensitivity at 24H, 48H, and 72H.

Median Vancomycin levels were consistently higher in patients infected with resistant organisms, which reflects clinical decisions to increase dosages to overcome antimicrobial resistance. The spread of values increased at later timepoints, suggesting accumulation over time, particularly in patients with impaired clearance. Additionally, extreme outliers were visible at 48 and 72 hours, emphasizing the need for individualized monitoring to minimize the risk of nephrotoxicity from excessive exposure.

3.4. Impact of Nephrotoxic Drugs on Kidney Function

To assess the effect of nephrotoxic drug exposure on kidney function, we examined the relationship between selected medications and serum creatinine (SCr), a key biomarker for renal performance. An SCr value above 1.7 mg/dL was used to define impaired kidney function. This threshold is clinically relevant, as values exceeding 1.7 mg/dL generally indicate at least moderate renal impairment, which may warrant altered dosing or close monitoring in hospital settings. The analysis focused on the following medications, included as binary (Yes/No) indicators: angiotensin-converting enzyme inhibitors (ACEI), loop diuretics, vasopressors, angiotensin II receptor blockers (ARB), non-steroidal anti-inflammatory drugs (NSAIDs), aminoglycosides, and Piperacillin/Tazobactam (PipTaz). We first examined the frequency of nephrotoxic drug use, as shown in Figure 4, followed by a multi-dimensional alluvial diagram (Figure 5) that links drug use, gender, and SCr level. Both plots were produced using base R and the `ggalluvial` package. The complete code for both visualizations is listed in Appendix 6.

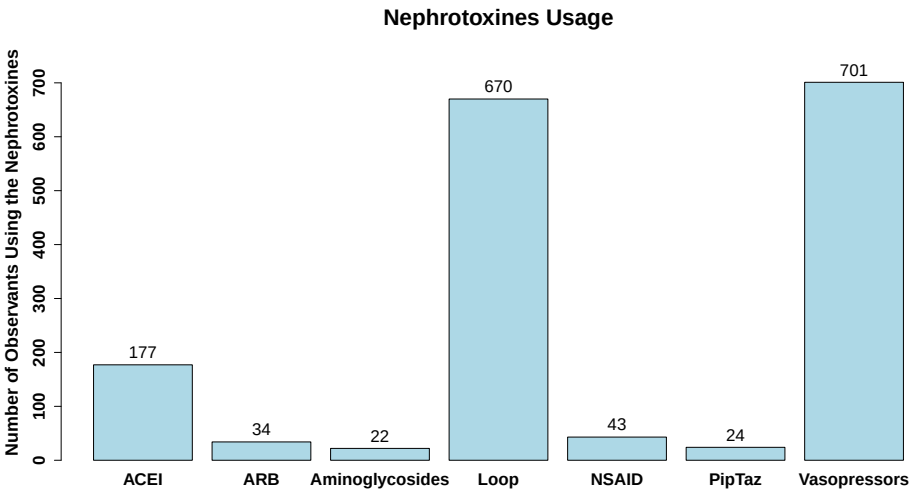

Figure 4: Frequency of Nephrotoxic Drug Usage

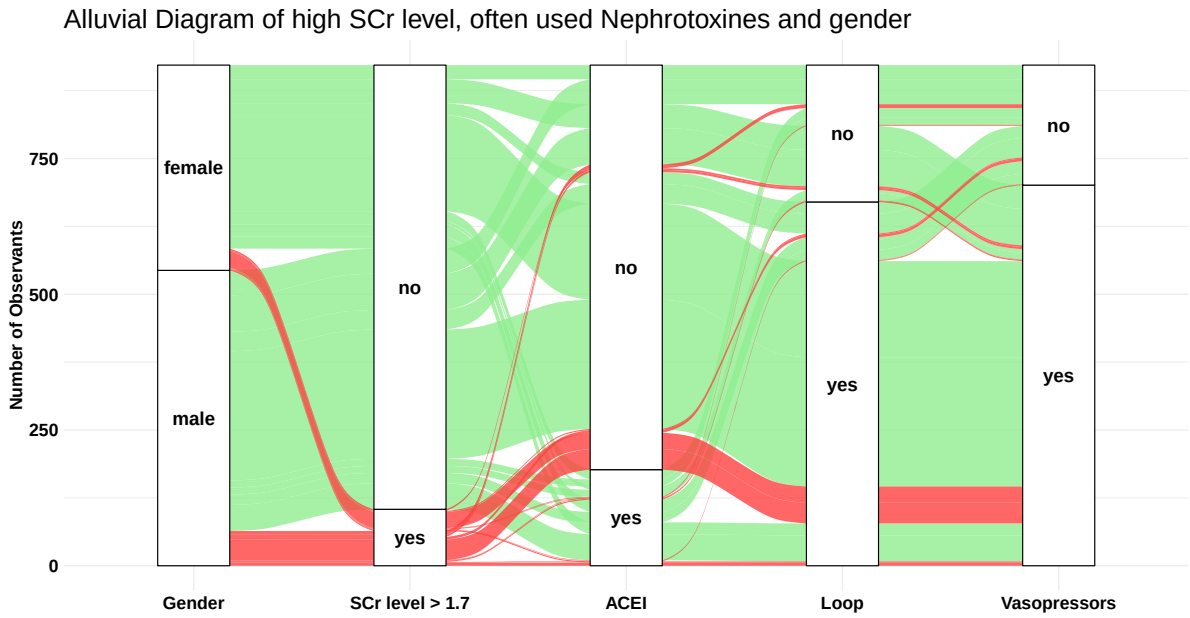

Figure 5: Alluvial Diagram of High SCr Level, Nephrotoxic Drug Use, and Gender. Red flows indicate patients with SCr > 1.7 mg/dL.

The alluvial diagram in Figure 5 visualizes the flow of patient subgroups across five variables: Gender, SCr level (>1.7 or not), ACEI use, Loop diuretic use, and Vasopressor use. Each vertical stratum represents one variable, while the flows (alluvia) show how patients transition from one category to another across these dimensions.

The color of each flow reflects the patient's SCr status—red flows represent patients with high SCr (>1.7 mg/dL), while green flows represent those with normal levels. This enables a quick visual assessment of which combinations are associated with impaired renal function. For example, several red flows can be observed passing through "no ACEI" and "yes Loop" or "yes Vasopressors," suggesting that patients not on ACEI but receiving Loop diuretics or Vasopressors were more likely to have elevated SCr. Gender differences are also noticeable, as more male patients appear in pathways that lead to high SCr values.

Loop diuretics and vasopressors were the most frequently administered nephrotoxic agents. A substantial number of patients with elevated serum creatinine had been exposed to these drugs, especially among male patients. Interestingly, ACEI use was less common in patients with impaired kidney function, possibly reflecting cautious prescribing practices due to known renal side effects. Overall, the diagram offers a multidimen-

sional view of risk factors for nephrotoxicity by combining medication use, gender, and renal outcomes.

4. Conclusion

The aim of this report was to explore clinical patterns in Vancomycin therapy using data visualization, with a particular focus on illness severity, renal function, bacterial resistance, and nephrotoxic drug exposure. By leveraging visual techniques, we sought to uncover relationships that could inform safer and more personalized treatment strategies in critically ill patients.

The results show that illness severity—as measured by SAPS and SOFA scores—has a clear impact on both kidney function and Vancomycin serum levels. Patients in higher severity categories demonstrated lower eGFR values over time and elevated Vancomycin concentrations, reflecting impaired renal clearance. SAPS-based stratification offered more distinct group separations compared to SOFA.

Infections with resistant bacteria were associated with higher median Vancomycin concentrations, suggesting that clinical teams responded by increasing dosage in such cases. However, the wider distributions and visible outliers at later timepoints highlight the need for close therapeutic drug monitoring to avoid toxicity.

Finally, the use of nephrotoxic medications—particularly loop diuretics and vasopressors—was common among patients with elevated serum creatinine levels. The alluvial diagram provided a nuanced understanding of the interplay between gender, medication exposure, and renal impairment.

In summary, the visual analyses confirmed several clinically relevant patterns in Vancomycin use and renal outcomes. The findings emphasize the importance of individualized therapy, particularly in patients with severe illness or those receiving nephrotoxic drugs. Visualization proved to be a powerful tool for uncovering these patterns and supporting data-driven clinical decision-making.

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A. Appendix

A.1. R Code for Visualizations

The following listings show the R code used to generate the plots in the analysis section.

The following R code in the listing 1 loads the R packages used.

Listing 1: Required R Packages for Vancomycin Data Visualization

```
# Required packages
library(ggplot2)          # For data visualization
library(reshape2)         # For data reshaping (melt, cast)
library(alluvial)         # For basic alluvial diagrams
library(ggalluvial)       # For advanced alluvial plots with
   ggplot2
library(dplyr)            # For data manipulation
library(patchwork)        # For combining ggplot2 plots
library(RColorBrewer)     # For colorblind-friendly palettes
library(tidyr)            # For pivot_longer and other
   tidying functions
```

R-Code for Figure 1

The code for generating Figure 1 can be found in Listing 2.

Listing 2: Line and Ribbon Plot of eGFR Change Over Time by SAPS/SOFA Categories

```
df_eGFR_long <- df_clean %>%
  select(SAPS_Category, SOFA_Category, eGFRStart, eGFR24,
     eGFR48, eGFR72, eGFREnd) %>%
  pivot_longer(cols = starts_with("eGFR"), names_to =
     "Timepoint", values_to = "eGFR") %>%
  mutate(Timepoint = factor(Timepoint, levels =
     c("eGFRStart", "eGFR24", "eGFR48", "eGFR72",
     "eGFREnd")))

saps_eGFR_summary <- df_eGFR_long %>%
```

```
  group_by(SAPS_Category, Timepoint) %>%
  summarise(
    median_eGFR = median(eGFR, na.rm = TRUE),
    Q1 = quantile(eGFR, 0.25, na.rm = TRUE),
    Q3 = quantile(eGFR, 0.75, na.rm = TRUE)
  )

p_saps <- ggplot(saps_eGFR_summary, aes(x = Timepoint, y =
   median_eGFR, group = SAPS_Category, color =
   SAPS_Category)) +
  geom_hline(yintercept = 90, linetype = "dashed", color =
     "black", size = 1) +
  geom_line(size = 1.2) +
  geom_ribbon(aes(ymin = Q1, ymax = Q3, fill =
     SAPS_Category), alpha = 0.2, color = NA) +
  labs(
    title = "eGFR␣Change␣Over␣Time␣(SAPS␣Categories)",
    x = "Timepoint",
    y = "eGFR␣(mL/min/1.73␣m²)"
  ) +
  theme_minimal()
```

R-Code for Figure 2

The code for generating Figure 2 can be found in Listing 3.

Listing 3: Boxplots of Vancomycin Concentrations by SAPS and SOFA Severity

```
saps_quantiles <- quantile(df_clean$SAPS, probs = c(1/3,
   2/3), na.rm = TRUE)
sofa_quantiles <- quantile(df_clean$SOFA, probs = c(1/3,
   2/3), na.rm = TRUE)

df_clean <- df_clean %>%
  mutate(
    SAPS_Category = factor(case_when(
```

```
      SAPS <= saps_quantiles[1] ~ "Low",
      SAPS > saps_quantiles[1] & SAPS <= saps_quantiles[2]
         ~ "Medium",
      SAPS > saps_quantiles[2] ~ "High"
    ), levels = c("Low", "Medium", "High")),
    SOFA_Category = factor(case_when(
      SOFA <= sofa_quantiles[1] ~ "Low",
      SOFA > sofa_quantiles[1] & SOFA <= sofa_quantiles[2]
         ~ "Medium",
      SOFA > sofa_quantiles[2] ~ "High"
    ), levels = c("Low", "Medium", "High"))
  )

format_boxplot_with_median <- function(data, x_var, y_var,
   fill_var, title, colors, y_limits = NULL) {
  medians <- data %>% group_by(!!sym(x_var)) %>%
    summarise(median_value = median(!!sym(y_var), na.rm =
       TRUE))

  p <- ggplot(data, aes(x = !!sym(x_var), y = !!sym(y_var),
     fill = !!sym(fill_var))) +
    geom_boxplot(coef = 1.5, width = 0.6, fatten = 1.2) +
    geom_text(data = medians, aes(x = !!sym(x_var), y =
       median_value, label = round(median_value, 1)),
              vjust = -0.5, size = 5, fontface = "bold") +
    labs(title = title, x = "Severity␣Category␣by␣
       Quantile", y = "Vancomycin␣Concentration␣(mg/L)") +
    theme_minimal() +
    scale_fill_manual(values = colors)

  if (!is.null(y_limits)) {
    p <- p + scale_y_continuous(limits = y_limits)
  }
  return(p)
}
```

```
p_c24_saps <- format_boxplot_with_median(df_clean,
   "SAPS_Category", "C24", "SAPS_Category", "C24␣Across␣
   SAPS␣Severity", saps_colors)
p_c24_sofa <- format_boxplot_with_median(df_clean,
   "SOFA_Category", "C24", "SOFA_Category", "C24␣Across␣
   SOFA␣Severity", sofa_colors)
p_c48_saps <- format_boxplot_with_median(df_clean,
   "SAPS_Category", "C48", "SAPS_Category", "C48␣Across␣
   SAPS␣Severity", saps_colors)
p_c48_sofa <- format_boxplot_with_median(df_clean,
   "SOFA_Category", "C48", "SOFA_Category", "C48␣Across␣
   SOFA␣Severity", sofa_colors)
p_c72_saps <- format_boxplot_with_median(df_clean,
   "SAPS_Category", "C72", "SAPS_Category", "C72␣Across␣
   SAPS␣Severity", saps_colors)
p_c72_sofa <- format_boxplot_with_median(df_clean,
   "SOFA_Category", "C72", "SOFA_Category", "C72␣Across␣
   SOFA␣Severity", sofa_colors)

final_plot <- (p_c24_saps | p_c48_saps | p_c72_saps) /
   (p_c24_sofa | p_c48_sofa | p_c72_sofa)
```

R-Code for Figure 3

The code for generating Figure 3 can be found in Listing 4.

Listing 4: Violin and Boxplot of Vancomycin Serum Levels by Bacterial Sensitivity

```
df_selected <- df_clean %>% select(Culture, C24, C48, C72)
df_selected$Culture <- as.factor(df_selected$Culture)
df_selected <- df_selected %>%
  filter_all(all_vars(!is.na(.))) %>%
  filter_all(all_vars(!is.infinite(.)))
```

```
df_long <- melt(df_selected, id.vars = "Culture",
   variable.name = "Timepoint", value.name =
   "Vancomycin_Level")

violin_plot <- ggplot(df_long, aes(x = Timepoint, y =
   Vancomycin_Level, fill = Culture)) +
  geom_violin(trim = FALSE, alpha = 0.7) +
  geom_boxplot(width = 0.2, position = position_dodge(0.9),
     outlier.shape = NA, alpha = 0.7) +
  scale_fill_manual(values = c("red", "green")) +
  scale_x_discrete(labels = c("C24" = "24H", "C48" = "48H",
     "C72" = "72H")) +
  labs(title = "Vancomycin␣Serum␣Levels␣by␣Bacterial␣
     Sensitivity",
       x = "Timepoint␣After␣Start␣of␣Therapy",
       y = "Vancomycin␣Serum␣Level␣(mg/L)",
       fill = "Bacterial␣Sensitivity") +
  theme_minimal()
```

R-Code for Figure 4

The code for creating the figure 4 can be found in listing 5.

Listing 5: Barplot and Alluvial Diagram of Nephrotoxic Drug Usage and SCr Status

```
# Nephrotoxine
sum(dat$ACEI == "yes")                # 177
sum(dat$ARB == "yes")                 # 34
sum(dat$Aminoglycosides == "yes")     # 22
sum(dat$Loop == "yes")                # 670
sum(dat$NSAID == "yes")               # 43
sum(dat$PipTaz == "yes")              # 24
sum(dat$Vasopressors == "yes")        # 701

pdf(file="nephrotoxines_usage.pdf", width=15, height=8)
old.par <- par(mar = c(5,8,4,2)+0.1)
```

```r
values <- c(177, 34, 22, 670, 43, 24, 701)

barplot_obj <- barplot(
  values,
  names.arg = c("ACEI", "ARB", "Aminoglycosides", "Loop",
     "NSAID", "PipTaz", "Vasopressors"),
  col = "lightblue",
  main = "Nephrotoxines Usage",
  ylab = "Number of Observants Using the Nephrotoxines",
  ylim = c(0, max(values) * 1.1),

  # Text sizes
  cex.names = 1.5,        # x-axis labels
  cex.axis = 1.5,         # y-axis tick labels
  cex.lab = 1.6,          # axis titles
  cex.main = 2.0,         # main title

  # Text styles
  font.main = 2,          # bold title
  font.lab = 2,           # bold axis labels
  font.axis = 2,          # bold axis ticks

  # Colors
  col.lab = "black",      # axis labels
  col.axis = "black",     # tick labels
  col.main = "black"      # main title
)
```

R-Code for Figure 5

The code for creating the figure 5 can be found in listing 6.

Listing 6: Alluvial Diagram of Nephrotoxic Drug Usage and SCr Status

```
# Create kidney_failure variable
dat$kidney_failure <- "no"
dat$kidney_failure[dat$SCrEnd > 1.7 | dat$SCr72 > 1.7 |
                   dat$SCr48 > 1.7 | dat$SCr24 > 1.7 |
                   dat$SCrStart > 1.7] <- "yes"

# Aggregate alluvial data
dat_diseases <- dat[, c("Gender", "ACEI", "Loop",
   "Vasopressors", "kidney_failure")]
dat_diseases$people <- 1
dat_diseases_agg <- aggregate(people ~ ., dat_diseases, FUN
   = sum)

# Alluvial plot
ggplot(dat_diseases_agg,
       aes(axis1 = Gender, axis2 = kidney_failure, axis3 =
          ACEI,
           axis4 = Loop, axis5 = Vasopressors, y = people))
              +
  geom_alluvium(aes(fill = kidney_failure), alpha = 0.8) +
  geom_stratum() +
  geom_text(stat = "stratum",
            aes(label = after_stat(stratum)),
            size = 6,
            fontface = "bold",
            color = "black") +
  scale_fill_manual(values = c("no" = "lightgreen", "yes" =
     "brown1")) +
  scale_x_discrete(limits = c("Gender", "SCr␣level␣>␣1.7",
     "ACEI", "Loop", "Vasopressors")) +
```

```
labs(title = "Alluvial Diagram of high SCr level, often
   used Nephrotoxines and gender",
     x = "",
     y = "Number of Observants") +
theme_minimal() +
theme(
  legend.position = "none",
  text = element_text(size = 20),
  axis.text.x = element_text(size = 16, face = "bold",
     color = "black"),
  axis.text.y = element_text(size = 16, face = "bold",
     color = "black"),
  axis.title.x = element_text(size = 16, face = "bold",
     color = "black"),
  axis.title.y = element_text(size = 16, face = "bold",
     color = "black")
)
```